Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/10/2009
PI Name:	Young, Laurence R. Sc.D.		
Project Title:	Sensorimotor Displays and Controls to Enhance the Safety of Human/Machine Cooperation During Lunar Landing		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	8
Comments:	Deceased as of August 2021.		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	07/01/2008	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Oman, Charles (Massachusetts Institute of Technology) Wood, Scott (NASA JSC) Duda, Kevin (The Charles Stark Draper Laboratory, Inc.) Estrada, Arthur (US Army Aeromedical Research Laboratory)		
Grant/Contract No.:	NCC 9-58-SA01604		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Lunar landing depends on the selection and identification of an appropriate location that is level and free of hazards, along with a stable controlled descent to the surface. During crewed landings, astronauts are expected to interact with automated systems, based upon improved terrain maps and sensor updates, to perform tasks such as manual re-designation of landing point, adjustment of descent trajectory or direct manual control. However, sensorimotor limitations, both vestibular and visual, are likely to interfere with performance and safety. This integrated project examines the nature of the anticipated spatial disorientation and terrain perception limits as they affect the transition from automatic to manual control, and develops advanced display countermeasures to overcome these limitations. There are four specific aims investigated in this multi-institution effort: (1) Examine the nature of anticipated sensorimotor difficulties (e.g., spatial disorientation, limits on terrain perception) as they affect the transition from automatic to manual control, (2) Develop and evaluate advanced display countermeasures for enhancing situation and terrain awareness and for overcoming performance limitations caused by reduced visibility associated with lunar lighting, terrain reflectivity and the absence of atmosphere utilizing Draper Laboratory's fixed-base lunar lander cockpit simulator for full human-in-the-loop evaluation. (3) Evaluate the effectiveness of the cockpit displays during human-in-the-loop manual control in the JSC TTS during "critical" and "hover" tasks testing the tilt-translation and tilt-gain illusions of altered acceleration sensitivity as it applies to lunar gravity following a period of weightlessness, and (4) Perform a series of evaluations of the displays using the U.S. Army Aeromedical Research Laboratory's six-degree-of-freedom helicopter simulator as a lunar landing analog for replicating lunar lighting and the various parameters associated with dust "brownout" conditions. In our first year of work, we have made significant progress in Aims 1 and 2. We have analyzed a set of candidate Altair landing trajectories using a physiologically-based model of human spatial orientation and concluded that there is a likely risk of pilot perceptions differing from actual vehicle state. The analysis does not yet account for how visual information from cockpit displays (e.g., an attitude indicator) or outside views (e.g., the horizon) contribute or counteract this misperception. We have also extensively reviewed the reports from the Apollo landings for descriptions of situations where environmental conditions contributed to a loss of spatial awareness and geographical disorientation. We have also made connections with a group at the NASA-Ames Research Center who are conducting a study of the handling qualities of the lunar lander with the goal of sharing relevant research results. In Aim 2, we have completed a review of the relevant literature on helicopter and vertical take-off/landing (VTOL) control modes and primary flight displays. This included a visit to the Sikorsky Aircraft Co in CT to discuss their work on the DARPA Sandblaster project, which is developing displays for helicopter landing in brownout/whiteout conditions. Following this review, we have begun developing the first set of prototype cockpit displays and controls for the Draper lab simulator. These displays are focused on the final hover and descent tasks, providing information on lander attitude, fuel usage and geographical awareness. The first displays have also been tested in a simple MATLAB/Simulink environment and will soon be ready to integrate into the full simulation. The team visited the NASA Johnson Space Center to review Dr. Wood's Tilt-Translation Sled (TTS) and discuss the integration of the Draper hardware and software into the TTS. The hardware and software interfaces have been determined and we are now in the process of finalizing the equipment to achieve some commonality between Draper and JSC simulations. The Aim 3 experiment protocol will be ready for submission to the JSC Institutional Review Board by the end of May. The groundwork for the Aim 4 experiments, which are not scheduled until Years 3-4, is being started as the US Army will be modifying their 6-DOF helicopter simulation to allow changes in the dust simulation. This will enable us to alter dust properties to match those of lunar regolith. For the second year, we will continue our work in Aim 1 examining the effects of lighting and dust on spatial orientation and manual control during landing. For Aim 2, we will finish the development of a set of prototype cockpit displays for the Draper simulator. Experiments using these displays to land on the lunar surface will be carried out in the Draper simulator. Flight and landing performance, situation awareness and pilot workload will be evaluated. We will update the landing trajectory analysis and display experiments as more information about the flight dynamics of the actual Altair vehicle are released. In the summer of 2009, after JSC-IRB approval, we will complete the installation of the Draper hardware and software into the JSC Tilt-Translation Sled. Once complete, we will begin a pilot experiment to verify that tilt-translation illusions can be generated with the system. The main experiment examining the effect of sensory discord on landing performance will be carried out as described in the proposal. We anticipate that work on the US Army Aerospace Research Lab simulator will begin and be completed in Year 2.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Human rating requirements currently mandate the capability for a "graceful reversion" from automated to manual control for spaceflight control systems. Our goal is to determine the limits of human performance under likely landing conditions that may cause spatial disorientation. Appropriate roles can thereby be selected for humans and automated systems. This proposed project will contribute to a better understanding of visual and vestibular conditions contributing to spatial disorientation during landing and the resulting effects on human manual control. We will have demonstrated display and control system interfaces to reduce pilot workload, improve situation awareness, and mitigate spatial disorientation to ensure a safe crewed lunar landing. Finally, the work may also have terrestrial applications in mitigating the risk of helicopter accidents by suggesting new techniques to address problems associated with brownout during landing.
	Progress for Aim 1: We have extracted relevant parameters from candidate trajectories within the NASA Autonomous Landing and Hazard Avoidance Technology (ALHAT) Project tradespace. These parameters were used as inputs to the Observer Model, which is a physiologically based model of spatial orientation to estimate perceived orientation of the astronaut within the lunar landing vehicle from the initiation of the braking burn to descent from lunar orbit all the way through touchdown on the surface of the moon. The Apollo landings as well as target landing points for future lunar missions have been extensively reviewed for instances and environmental conditions that may lead to issues surrounding terrain awareness and geographic disorientation. Dr. Oman visited the NASA-Ames VMS in January to discuss the handling qualities project led by Dr. Karl Bilimoria. The discussions and simulator session indicate that augmentation will be needed for the lander. Progress for Aim 2: The three graduate student RAs have been badged and given offices at the Draper Labs. Relevant literature on helicopter and vertical take-off and land (VTOL) control modes as well was primary flight and situation awareness displays were

	reviewed. This included a visit to Sikorsky Aircraft in Stratford, CT to discuss the displays developed for the DARPA Sandblaster Project. This review, combined with recommendations in appropriate MIL-STD and SAE ARP documents were used to generate designs for lunar lander control modes and to define the information requirements for flight displays.
Task Progress:	Prototype control modes and primary flight displays have been implemented in MATLAB/Simulink for initial evaluation. These control modes receive inputs from a joystick connected to the computer and the resulting flight dynamics drive elements of the flight displays. Representative dynamics of the Altair lander were used for analysis based on the LDAC-1 vehicle parameters, which are implemented in the Draper cockpit simulator and can be flown manually. Progress for Aim 3: Dr. Wood hosted a team visit at JSC in February to view the TTS facilities. Two manual control tasks are being integrated and the real-time LabView software interface has been modified to used shared variables to enable the Draper landing displays and manual control task to be performed on the sled. The protocol for the Aim 3 experiments is being prepared and will be submitted to the JSC-IRB by the end of May 2009. Progress for Aim 4: Dr. Estrada was able to secure a solicitation from the US Army to make modifications to their helicopter simulation which will enable us to manipulate the dust characteristics for the Aim 4 experiments, which will occur in Years 3-4. The costs of the change will be assumed entirely by the US Army Helicopter Project Office. Dr. Estrada has also provided subject matter expertise on aeromedical/physiological issues, hover flight techniques, cockpit displays, and situational awareness.
Bibliography Type:	Description: (Last Updated: 01/21/2026)
Abstracts for Journals and Proceedings	Duda KR, Young LR, Oman CM, Liu AM, Stimpson AJ, Clark TK. "Evaluation of sensorimotor performance during lunar landing." 80th Annual Scientific Meeting of the Aerospace Medical Association, Los Angeles, CA, May 4-7, 2009. Aviat Space Env Med. 2009 Mar;80(3):230. , Mar-2009
Abstracts for Journals and Proceedings	Young LR, Duda KR, Oman CM, Liu AM, Stimpson AJ, Clark TK. "Critical factors affecting lunar landing supervisory control performance." 60th International Astronautical Congress, Daejeon, Republic of Korea, October 12-16, 2009. 60th International Astronautical Congress, Abstract Book, October 2009. , Oct-2009